Introduction to Programming

Lecture 30

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In Today's Lecture

- References
- Differences Between References and Pointers

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Reference

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&

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int & i ; 'i' is a reference to an integer
int * i ; 'i' is a pointer to an integer

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```
int i ;
int & j = i ;
```

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Example 1

```
main ( )
{
   int i ;
   int & j = i ;
   i = 123 ;
   cout << i << endl ;
   cout << j << endl ;
   i++ ;
   cout << i << endl ;
   cout << j ;
}
```

Output

```
123
123

124
124
```

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Example 2

```
void swap ( int x , int y )
{
  int temp = x ;
  x = y ;
  y = temp ;
}
```

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Example 2

```
main ( )
{
  int x , y ;
  x = 10 ;
  y = 20 ;
  swap ( x , y ) ;
  cout << x << y ;
}
```

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Example 2

```
main ( )
{
  int x , y ;
  x = 10 ; y = 20 ;
  swap ( & x , & y ) ;
  cout << x << y ;
}
```

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```
swap ( int * i , int * j ) ;
```

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Example 2

```
void swap ( int * i , int * j )
{
  int * temp = i ;
  * i = * j ;
  * j = * temp ;
}
```

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Example 3

```
void swap ( int & i , int & j )
{
  int temp = i ;
  i = j ;
  j = temp ;
}
```

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Example 3

```
main ( )
{
  int x , y ;
  x = 10 ;
  y = 20 ;
  swap ( x , y ) ;
  cout << x << y ;
}
```

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const

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Example 4

```
struct b
{
   int serno ;
   char text [ 1000 ] ;   // A lot of chars
} b = { 123, "This is a BIG structure" } ;

void valfunc ( b ) ;        // Call by value

void ptrfunc ( * b ) ;     // Call by pointer

void reffunc ( & b ) ;   // Call by reference
```

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Example 4

```
void main ( )
{
   valfunc ( b ) ;        // Passing the variable itself
   ptrfunc ( & b ) ;      // Passing the address of the variable
   reffunc ( b ) ;        // Passing a reference to the variable
}
```

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Example 4

```
struct b
{
   int serno ;
   char text [ 1000 ] ;   // A lot of chars
} b = { 123 , "This is a BIG structure" } ;

void valfunc ( b ) ;                  // Call by value

void ptrfunc ( const * b ) ;    // Call by pointer

void reffunc ( const & b ) ;   // Call by reference
```

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Difference Between References and Pointers

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```
main ( )
{
  int i ;
  int & refi = i ;
  cout << "The address of the
                reference is " << & refi ;
  cout << & i ;
}
```

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Dangling Reference

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References as Return Values

```
int mynum = 0 ;    // Global variable
int & num ( )
{
   return mynum ;
}
void main ( )
{
   int i ;
   mynum = 100 ;
   i = num ( ) ;
   num ( ) = 200 ;
}
```

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```
int & i ;
double & f ;
char & c ;
```

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```
a = b = c ;
```

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$$a = b;$$

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```
cout << "The values of the
        integer is  "<< i << endl ;
```

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